SevaOne: High-Level Architecture (Web)

Current implementation of the web platform, drawn from the repository and the running stack.

Service · Request · Event or callback · Data store · Runs on a separate server

1 Clients

Citizen web portal
React 19 · Vite · TanStack Router
Tailwind 4 · Zustand

Operator console
Assisted / kiosk mode
Part of the same React app

Admin panel
Government admin console
React 19 · analytics, catalogue

HTTPS

2 Edge and security

NGINX reverse proxy
TLS 1.2 / 1.3 · security headers (CSP, frame deny)

citizen routes

operator + department routes

3 Gateway and identity

WSO2 API Manager 4.7: citizen
API product sevaone-citizen · gateway :8243
Citizen routes, API key and gateway secret

WSO2 API Manager 4.7: department
API product sevaone · gateway :8243
Operator and department routes

WSO2 Identity Server 7.3
SCIM2 · OAuth2 JWT-bearer (RFC 7523)
Citizen, operator and admin issuers

forwarded + secret

forwarded + secret

SCIM2 + JWT

department callbacks

4 Application services

AI services
GPU server, own network

sevaone-rag: agentic RAG
vLLM language model
FAISS or Qdrant index
Bhashini for Kannada

ai-admin and ai-monitor
over SSH tunnels

ai-chat proxy

.NET 10 backend (SevaOne.Services)
Dapper + Npgsql · FluentValidation · Serilog · OpenTelemetry

- **Services** — Controllers and middleware
- **Business** — Apply, track, department handoff
- **Infrastructure** — Postgres, WSO2, DigiLocker, SMS clients
- **Domain** — Models and error codes

OutboxDispatcher publishes events to Kafka · Flyway migrations at start-up

Npgsql · functions

sessions · cache

outbox events

outbound calls and redirects

5 Data and streaming

PostgreSQL 18
6 schemas: discovery, portal, orchestration, status, audit, notification
Function-only access

Redis 8
Sessions
Token-check cache
Data-protection keys

Apache Kafka 4.3
Transactional outbox events

6 External systems

DigiLocker
CeG gateway
Aadhaar OTP, eAadhaar

SMS gateway
One-time passwords
Mobile Seva

Public catalogue sites
Sakala · ESCOM
Seva Sindhu catalogue

Department systems
Nadakacheri (Revenue)
Department Hub · AJSK